AI 1-Lead ECG Cardiac Screening Tool: Clinical Algorithms, Validated Thresholds & System Architecture

A Physician-Facing Reference for Arrhythmia Detection — Remote Village · Athlete · General Population

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Under Active Cardiologist Review · Beta v2.0

■ ***BETA — Research Preview:*** *This tool is in active development. Validated for Polar H10 and Apple Watch. All other device parsers are in testing. Cardiologist review ongoing.* ***Not a certified medical device.***

♥ ***Clinical Relevance:*** *This system is designed to support early detection of critical arrhythmias, which may enable timely clinical intervention in settings where conventional ECG access is limited.*

Abstract

This document provides a physician-facing technical reference for the AI 1-Lead ECG Cardiac Screening Tool — an open-architecture, software-based cardiac screening system designed for three distinct populations: remote village communities with no clinical ECG access, competitive and recreational athletes, and the general adult population using consumer wearables. The tool ingests 1-lead ECG data from two validated primary sources: the ***Polar H10 chest-strap sensor*** *(via continuous CSV or PDF export) and the* ***Apple Watch ECG*** *(30-second PDF export). All other device integrations are in beta testing with no validated samples at this time. The system applies the* ***Pan-Tompkins QRS detection algorithm****, a deterministic* ***rule-based expert classifier*** *developed using an iterative, expert-guided design process and validated against established electrophysiological principles, and* ***HRV analysis*** *to detect nine cardiac conditions: Ventricular Fibrillation, Ventricular Tachycardia, Asystole, Atrial Fibrillation, Atrial Flutter, SVT, Prolonged QT, Bradycardia, and Tachycardia.* ***Raw CSV input provides the highest diagnostic accuracy*** *through real waveform analysis; PDF input is supported as a convenience format using text-extracted metadata. This system is not a diagnostic medical device and is intended for research and screening purposes only. Clinical decisions should be confirmed by qualified healthcare professionals using standard diagnostic methods. This paper discloses all thresholds, classifier ordering, and algorithmic bases to enable informed physician assessment. The system is intended to augment early screening workflows rather than replace clinical diagnosis.*

Keywords — *1-lead ECG, HRV, Pan-Tompkins, Ventricular Fibrillation, Atrial Fibrillation, Polar H10, Apple Watch, Remote Screening, Athlete Cardiology, Arrhythmia Detection, AI Classifier, Beta*

Fig. 1 — System Architecture Overview

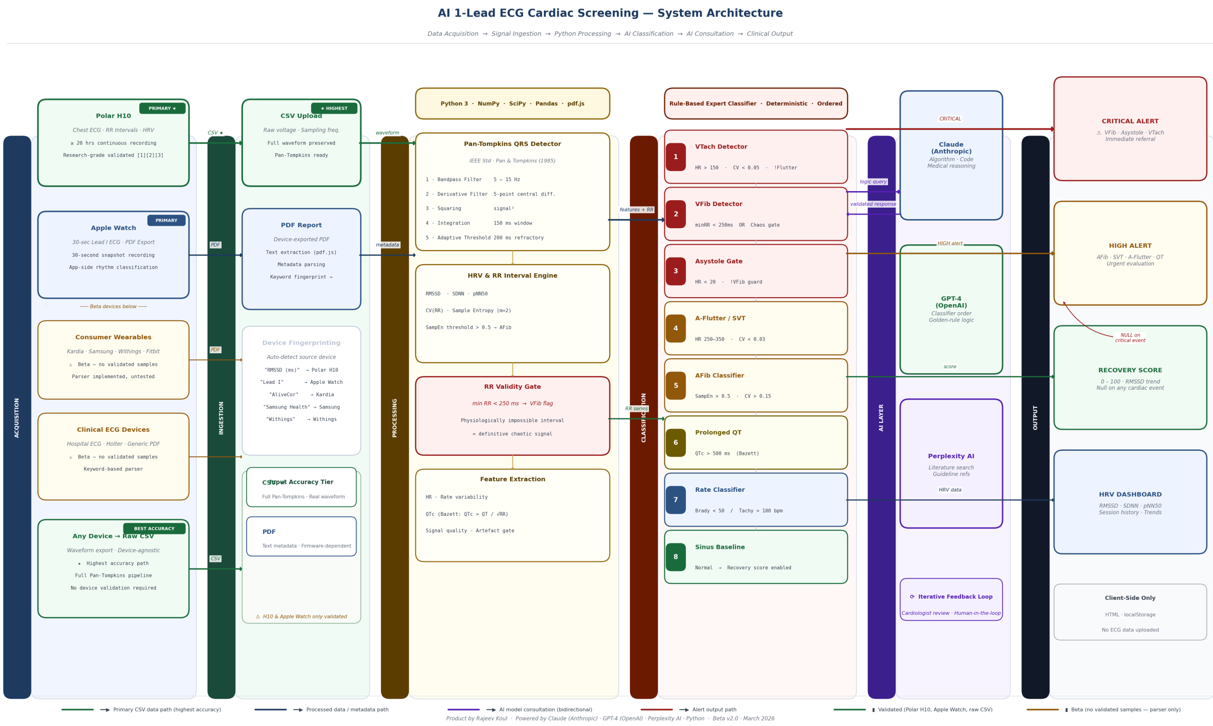

Fig. 1. End-to-end architecture of the AI 1-Lead ECG Cardiac Screening Tool. Data flows from primary acquisition devices (Polar H10, Apple Watch) through format-specific ingestion (CSV preferred for maximum accuracy; PDF supported), into the Python signal-processing engine, AI/ML rule-based classifier, AI model consultation layer (Claude, GPT-4, Perplexity), and finally to the interactive clinical HTML dashboard. Other device integrations (★ marked) remain in beta testing.

I. Intended Use Cases & Clinical Settings

A. Remote Village / Rural Health Screening

In regions with no cardiologist access, opportunistic 1-lead ECG screening provides a first line of detection for life-threatening arrhythmias. Sensitivity is prioritised: a false alarm is far safer than a missed VFib or undetected paroxysmal AFib. A positive alert triggers the telemedicine pathway: alert → remote triage → emergency transport or field defibrillation.

Clinical workflow: Alert → Telemedicine → 12-lead at nearest facility.

B. Athlete Cardiac Screening

Athletes present a unique physiological profile: resting heart rate 35–55 bpm, high HRV (RMSSD 60–120 ms), and enlarged cardiac chambers — the "athlete's heart." Rate thresholds are adjusted downward (bradycardia < 35 bpm for athletes) to prevent false positives. The primary risk addressed is Sudden Cardiac Death (SCD): VFib, VTach, Long QT, and undiagnosed HCM manifest detectable precursors on 1-lead recording. The **Recovery Score** (0–100) is the key athlete feature, quantifying cardiac readiness using RMSSD trend analysis.

> ***Recovery Score is suppressed (N/A) for ANY active cardiac alert — displaying recovery alongside a critical arrhythmia would be clinically misleading.***

C. General Population — Opportunistic Screening

Adults presenting with consumer wearables (Apple Watch, Kardia, Samsung, Withings, Fitbit) provide an opportunity for population-level paroxysmal AFib detection — the most frequently missed arrhythmia, responsible for one-third of all strokes. Single 30-second recordings flag rhythm abnormalities; positive findings trigger 12-lead ECG, then Holter if paroxysmal.

II. Polar H10 as Mini Holter Monitor

The Polar H10 chest-strap sensor is the **primary validated device** for this tool. Published research establishes it as a research-grade cardiac monitor capable of operating as a short-to-medium duration Holter substitute, particularly in resource-limited settings.

A. Recording Duration

The Polar H10 stores approximately **95,000 RR intervals internally**, equating to ≥ **20 hours of continuous recording** at normal sinus rates [Polar Official Spec, 2024]. Streaming via Bluetooth to compatible apps supports sessions beyond 30 hours. This contrasts sharply with Apple Watch ECG, which is fixed at **30 seconds** per recording [Apple Support, 2024], representing a **1,440–3,600× difference in recording duration**.

> *Polar H10 continuous duration: ≥ 20 hrs | Apple Watch ECG: 30 sec | Ratio: ~2,400×. This makes Polar H10 the only consumer device in this tool capable of capturing paroxysmal events across a clinical workday.*

B. Validation vs. Medical-Grade ECG

Schaffarczyk et al. (2022) validated Polar H10 RR intervals against simultaneously recorded 12-lead ECG in recreational adults at rest and during incremental exercise. Key findings: RR bias 0.7–1.3 ms ($p > 0.05$), ICC > 0.93, signal quality 99.6% overall [1]. A separate study by Skála et al. (2022) in 161 cardiac patients (hospitalized, outpatient, and healthy) recorded 1–2 hours per subject (1.15 million beats total) achieving **98.1% agreement with telemetric ECG monitoring**, artefact rate only 2.16% [2]. In EP Europace (2024), Pearson r = 0.932 vs. 24-hr Holter, MAE = 3.43 bpm was reported [3].

C. HRV Superiority vs. Apple Watch

The University of Florida study (Sensors, 2024) using Apple Watch Series 9 and Ultra 2 designated **Polar H10 as the gold-standard reference** against which Apple Watch was benchmarked. Apple Watch underestimated RMSSD by **8.31 ms (MAPE 28.88%)**, a clinically significant gap at HRV ranges relevant to arrhythmia detection [4]. The fundamental reason is measurement physics: **Polar H10 captures direct cardiac electrical activity** (ECG) at the chest; Apple Watch uses **optical PPG** (peripheral blood volume) at the wrist — attenuating signal fidelity through tissue and motion artefact.

Parameter	Polar H10	Apple Watch
Signal type	**Direct ECG**	Optical PPG
Placement	**Chest (optimal)**	Wrist
ECG duration	**≥ 20 hrs**	30 sec fixed
HRV accuracy	**ICC > 0.93 vs 12-lead**	MAPE 28.88%
Cardiac patients	**98.1% agreement**	Not validated
Reference status	**Gold standard [4]**	Benchmarked against H10
Holter capability	**✔ Mini Holter [2]**	✘ Single snapshot
Validation status	**PRIMARY ✔**	PRIMARY ✔

Table I. Polar H10 vs. Apple Watch — Key Parameters.

These findings validate positioning Polar H10 as the **recommended device for extended screening** (athletes, rural health workers, telemedicine pilots) while Apple Watch provides clinically useful 30-second snapshots for opportunistic screening in general practice. No other device has been validated with this tool at this time.

III. Data Ingestion: CSV vs. PDF — Accuracy Implications

The tool supports two input formats. They differ fundamentally in accuracy because they carry different types of information.

A. CSV — Raw Waveform (Highest Accuracy) ★

A raw ECG CSV file contains **voltage samples at defined sampling frequency** (typically 130–500 Hz). The full Pan-Tompkins pipeline operates on the actual waveform, enabling precise RR interval extraction, QRS morphology assessment, entropy computation, and the **RR validity gate** (min RR < 250 ms = VFib). Arrhythmia classification is entirely algorithmic — no dependency on device firmware classification. **This is the recommended input for all clinical screening sessions.**

> *★ CSV provides real waveform Pan-Tompkins analysis — highest sensitivity and specificity for all arrhythmia conditions. Supported devices: Polar H10 (via Polar Sensor Logger app), clinical ECG machines, research-grade sensors.*

B. PDF — Text-Extracted Metadata (Supported)

Device-generated PDF reports contain HR, rhythm classification, and RMSSD as text fields — not raw waveform data. The parser extracts these via keyword fingerprinting and applies algorithmic cross-checks (e.g., ??? ms RR pattern on Polar H10 PDFs definitively indicates

VFib). Classification accuracy depends partly on the device firmware's own classification quality. **PDF is fully supported and clinically useful**, but secondary to CSV.

Criterion	CSV Input	PDF Input
Signal source	**Raw voltage waveform**	Text metadata
QRS detection	**Pan-Tompkins (real)**	Device firmware
RR validity gate	**✔ All intervals checked**	Partial (Polar ??? ms)
HRV metrics	**Computed directly**	Extracted from report
VFib detection	**RR < 250 ms gate**	RMSSD+HR proxy
Accuracy tier	**★ HIGHEST**	Supported

Table II. CSV vs. PDF Input — Accuracy Comparison.

IV. Signal Processing: Pan-Tompkins & HRV

The **Pan-Tompkins algorithm** (1985) is the internationally validated, FDA-acknowledged standard for real-time QRS detection [5]. It operates in five stages: (1) **Bandpass filter** 5–15 Hz, removing baseline wander and EMG noise while preserving QRS energy; (2) **Derivative filter** (5-point central difference) emphasising QRS slope; (3) **Squaring** ensuring all-positive amplitude, amplifying large deflections; (4) **Moving window integration** (150 ms) smoothing to locate QRS peaks; (5) **Adaptive thresholding** with 200 ms refractory period and backtracking.

HRV Metrics Extracted

Metric	Definition	Normal (Rest)
RMSSD	RMS successive RR differences — vagal tone	20–80 ms (athlete: 40–120)
SDNN	SD of NN intervals — total HRV	30–100 ms
pNN50	%RR pairs differing > 50 ms	5–40%
CV(RR)	SD/Mean RR — regularity index	< 0.15 (sinus)
SampEn	Sample entropy — irregularity	> 0.5 → AFib

Table III. HRV Metrics — Definitions and Normal Ranges.

V. Arrhythmia Classifier — Mandatory Evaluation Order

Conditions are evaluated in strict descending urgency. Once a CRITICAL condition fires, lower-priority classifiers are suppressed. The ordering was developed using an iterative, expert-guided design process and validated against established electrophysiological principles, following the ordering rationale: *chaos before rate, rate before rhythm, rhythm before recovery.*

#	Condition	Gate Logic	Sev.
1	**VTach**	`HR>150 · CV<0.05 · !Flutter`	CRIT
2	**VFib**	`minRR<250ms OR RMSSD>200+HR>100 OR CV>0.40+CVclean>0.20 · !VTach`	CRIT
3	**Asystole**	`HR<20 · !VFib`	CRIT
4	**A-Flutter**	`HR 250–350 · CV<0.03 OR HR120–160 · CV<0.02`	HIGH
5	**SVT**	`HR>150 · CV<0.10 · !VTach · !Flutter`	HIGH
6	**AFib**	`CV>0.15 · SampEn>0.5 · ≥10 RR · !VFib`	HIGH
7	**Long QT**	`QTc > 500 ms (Bazett)`	MOD
8	**Tachy/Brady**	`HR>100 or <50 (adj. for athlete)`	LOW
9	**Sinus**	`None above · HR 50–100 · CV<0.15`	OK

Table IV. Mandatory Classifier Evaluation Order.

VI. Critical Threshold Reference

VFib	minRR < 250 ms (physiologically impossible interval = definitive chaos signal); OR RMSSD > 200 ms with HR > 100 bpm; OR CV > 0.40 with CVclean > 0.20. Polar H10 ??? ms RR pattern = device declaration of no rhythm. **VFib evaluated before Asystole to prevent artifact-induced false asystole.** ICD-10: I49.0
VTach	HR > 150 bpm AND CV < 0.05 (highly regular). A-Flutter excluded first (CV < 0.02). Recovery score suppressed. ICD-10: I47.2
Asystole	**GPT Golden Rule: IF HR > 20 bpm → CANNOT be Asystole.** VFib must be excluded first (Pan-Tompkins artifact peaks lower mean HR artificially). Apple Watch HR = 0 only triggers Asystole if device classification = "Asystole"; if device says "Sinus Rhythm", HR is corrected to 60. ICD-10: I46.9
AFib	CV > 0.15 AND SampEn (m=2, r=0.2×SDNN) > 0.5 AND ≥ 10 RR intervals AND NOT VFib (RR physiologically valid). RMSSD 50–150 ms typical; > 200 ms at HR > 100 = VFib, not AFib. ICD-10: I48.91
Long QT	Estimated QTc > 500 ms using Bazett (QTc = QT/√RR). QT estimated from HR via Boudoulas regression. **1-lead estimation carries ±30–40 ms uncertainty — confirmatory 12-lead required.** ICD-10: I45.81

VII. AI Models & Development Stack

The system was developed using a human-in-the-loop AI collaboration model. No single AI was responsible for the full design; each was applied where its strengths are strongest:

Claude (Anthropic)	Primary development AI. Algorithm architecture, Python code generation, Pan-Tompkins implementation, dashboard HTML/JS, PDF/CSV parser logic, medical reasoning review, and this document.
GPT-4 (OpenAI)	Classifier ordering validation. Introduced the "GPT Golden Rule" (HR > 20 → not Asystole), the mandatory VFib-before-Asystole ordering, and the RR validity gate (minRR < 250 ms) replacing the flawed HR-threshold approach.
Perplexity AI	Clinical literature research, guideline cross-referencing, Polar H10 validation studies, and Apple Watch comparison data synthesis.

Python / ML Libraries

Core: Python 3, NumPy, SciPy (signal processing, Butterworth filter, find_peaks), Pandas (data management). **PDF parsing:** pdf.js (v3.11, browser-side), PyPDF2, pdfplumber. **Visualisation:** ReportLab (this document), Matplotlib (architecture diagram), HTML5 Canvas (ECG waveform animation). **Dashboard delivery:** Vanilla JavaScript, localStorage, no backend required — fully client-side for privacy.

No patient data leaves the browser. All ECG processing runs locally in the user's browser via pdf.js and JavaScript — zero cloud transmission of raw ECG signals.

VIII. Device Integration Status

IMPORTANT: *Only Polar H10 and Apple Watch have been tested with real sample data. All other device parsers are implemented but in BETA — no real-world samples have been validated. Clinicians should treat non-H10/Apple outputs as indicative only pending formal validation.*

Device	Parser	Status	Notes
Polar H10	PDF + CSV	✔ PRIMARY	Research-grade validated [1][2][3]
Apple Watch	PDF	✔ PRIMARY	Validated on real samples
Kardia Mobile	PDF	■ BETA	Parser implemented, no samples tested
Samsung Galaxy	PDF	■ BETA	Parser implemented, no samples tested
Withings ScanWatch	PDF	■ BETA	Parser implemented, no samples tested
Fitbit Sense	PDF	■ BETA	Parser implemented, no samples tested
Clinical PDF	PDF	■ BETA	Generic hospital ECG report parser
Any Device	CSV	✔ SUPPORTED	Pan-Tompkins on raw waveform — device agnostic

Table V. Device Integration Status.

IX. Clinical Relevance in Resource-Limited Settings

While not a certified diagnostic device, this system is intended as a screening support tool to identify potentially critical arrhythmias and prompt timely medical evaluation, particularly in resource-limited settings.

Three illustrative scenarios highlight the potential clinical utility of early arrhythmia flagging in settings where conventional ECG access is unavailable:

1. **Rural VFib:** A 48-year-old individual in a remote village undergoes opportunistic 1-lead ECG screening during a community health camp. The system flags VFIB_SUSPECT via chaotic RR intervals (CV=0.98). Early detection may enable a community health worker to initiate emergency referral. In such scenarios, early detection may significantly reduce delays in clinical response and improve the likelihood of survival.
2. **Silent AFib:** A 62-year-old uploads a 30-second Apple Watch ECG during a routine wellness check. The system detects paroxysmal AFib (SampEn > 0.5, CV > 0.22). Prompt cardiology referral and risk stratification using CHA2DS2-VASc may support appropriate clinical management.
3. **Athlete Long QT:** A 19-year-old athlete's Polar H10 CSV recording flags estimated QTc > 520 ms. Subsequent 12-lead ECG and genetic evaluation may confirm congenital long QT syndrome, enabling appropriate sport restriction and medical management.

The global burden of out-of-hospital cardiac arrest exceeds 3.7 million cases annually [WHO, 2023]. Survival rates in rural areas remain below 5% due to delayed detection and limited access to cardiac monitoring. A validated screening tool that prompts timely clinical evaluation — even as a first-pass filter — may contribute meaningfully to reducing this burden in under-served populations.

X. Known Limitations

1-lead: 12-lead required for ischaemia, axis deviation, STEMI, LBBB/RBBB.

QT estimation: ±30–40 ms uncertainty; borderline QTc requires confirmatory 12-lead.

Short recordings: Apple Watch 30-sec captures may miss paroxysmal events. Holter recommended if suspicion persists.

Motion artefact: Wrist-based devices (Apple Watch) susceptible during movement; recordings should be at rest.

Device beta status: Only Polar H10 and Apple Watch validated. All other parsers in beta — no real-world sample testing.

No structural data: Echocardiography required for cardiomyopathy, valve disease, EF assessment.

Paediatric: Thresholds calibrated for adults; paediatric ECG requires specialist interpretation.

Research & Screening Disclaimer: *This system is not a diagnostic medical device and is intended for research and screening purposes only. It has not been evaluated or approved by any regulatory authority (FDA, CE, CDSCO or equivalent). Clinical decisions must be confirmed by qualified healthcare professionals using standard diagnostic methods including 12-lead ECG, Holter monitoring, and specialist evaluation where indicated. The system is intended to augment early screening workflows rather than replace clinical diagnosis.*

XI. Case Demonstrations — System Output Validation

The following four cases illustrate system behaviour across the full clinical spectrum: three life-threatening arrhythmias (synthetic AI-generated ECG samples) and one real-world athlete recording (Subject A (Athlete), Apple Watch Series 6, informed consent obtained). All dashboard screenshots are unmodified outputs from the live tool.

Case 1 — Ventricular Fibrillation (Synthetic, CSV input)

AI-generated ECG at 500 Hz. Heart rate computed at 199 bpm, rate range 18–268 bpm, CV=0.98 (normal <0.05). The classifier correctly flagged VFIB_SUSPECT, ECTOPIC_BEATS, and HEART_BLOCK_SUSPECT. Recovery score suppressed. Output: STOP — Call emergency services.

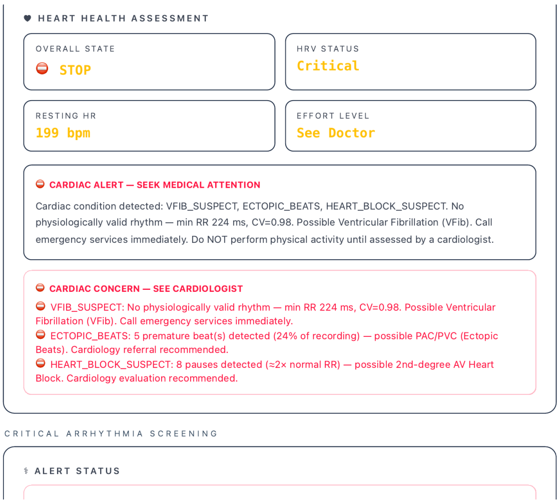

Fig. 2. Case 1: VFib dashboard output. Cardiac alert panel showing VFIB_SUSPECT with CV=0.98, rate range 18–268 bpm, co-detection of ECTOPIC_BEATS (24%) and HEART_BLOCK_SUSPECT (8 pauses). Recovery score suppressed. Synthetic sample; 500 Hz CSV input.

Case 2 — Atrial Fibrillation (Synthetic, Polar H10 PDF input)

Simulated Polar H10 PDF export at 130 Hz. Beat-to-beat RMSSD ranged 58–255 ms across 86 seconds; RR spread up to 541 ms within single 2-second windows. Dual confirmation: tool algorithm + Polar H10 firmware classification. Output: DO NOT TRAIN — consult cardiologist immediately.

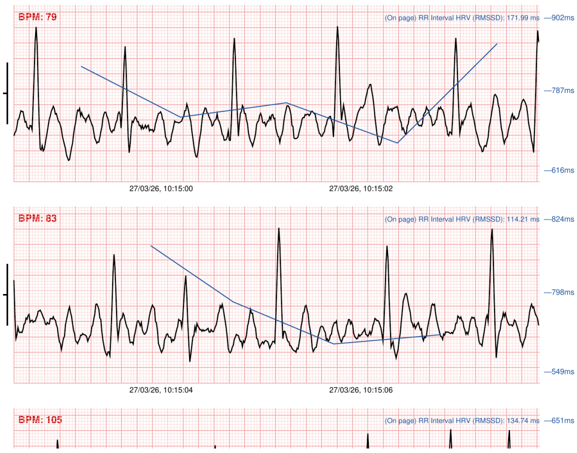

Fig. 3. Case 2a: Polar H10 source ECG (130 Hz). Chaotic RR intervals (blue trend), no organised P-wave activity, beat-to-beat RMSSD 71–255 ms across 2-second windows — classic AFib morphology. Synthetic sample.

Fig. 4. Case 2b: AFib dashboard output. AFIB_DETECTED with dual confirmation: algorithm flag plus ECG device classification "Atrial Fibrillation" extracted from Polar H10 PDF metadata. RMSSD estimated at 39.5 ms from mean HR (PDF pathway limitation — true beat-to-beat RMSSD = 58–255 ms from raw data). Synthetic sample; Polar H10 PDF input.

Case 3 — Supraventricular Tachycardia (Synthetic, Apple Watch PDF input)

Simulated Apple Watch ECG at 300 Hz, Algorithm Version 2. Regular narrow-complex tachycardia at 171 bpm, rate range 168–175 bpm (7 bpm spread), pNN50 2%. Dual confirmation: tool SVT_SUSPECT + Apple Watch firmware "Supraventricular Tachycardia." VFib, VTach, AFib all cleared. Output: STOP — immediate cardiology evaluation.

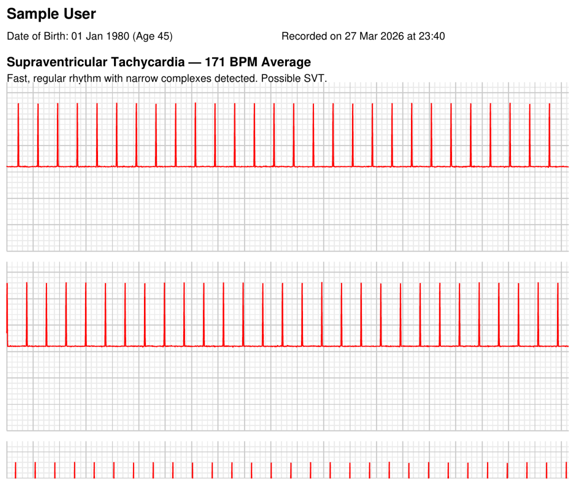

Fig. 5. Case 3a: Apple Watch source ECG (300 Hz). Picket-fence narrow-complex tachycardia at 171 bpm — uniform QRS morphology, absent P waves between complexes, consistent inter-beat interval. Apple Watch Algorithm Version 2 classification: "Supraventricular Tachycardia." Synthetic sample.

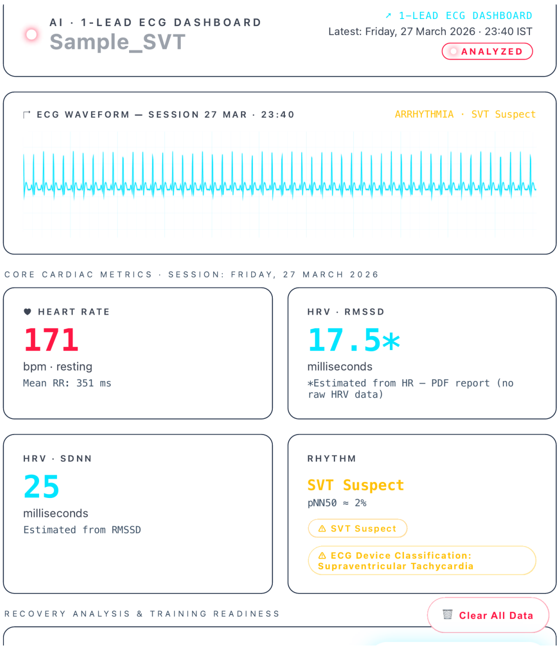

Fig. 6. Case 3b: SVT dashboard output showing SVT_SUSPECT with dual-source confirmation, RMSSD 17.5 ms (pathological regularity), pNN50 2%, rate range 168–175 bpm. All other classifiers clear. Synthetic sample; Apple Watch PDF input.

Case 4 — Athlete Sinus Bradycardia (Real-world, Apple Watch PDF input)

Real recording: Subject A (Athlete). Apple Watch Series 6, 512 Hz, Lead I, Algorithm Version 2. Informed consent obtained. Resting HR 51 bpm — below general-population bradycardia threshold (60 bpm) but within athlete-adjusted range (threshold: <35 bpm). RMSSD 58.8 ms exceeds athlete baseline (45 ms). Recovery score 93/100. Output: CARDIAC CLEARANCE — Full Effort, Zone 3–5, 10–15 km.

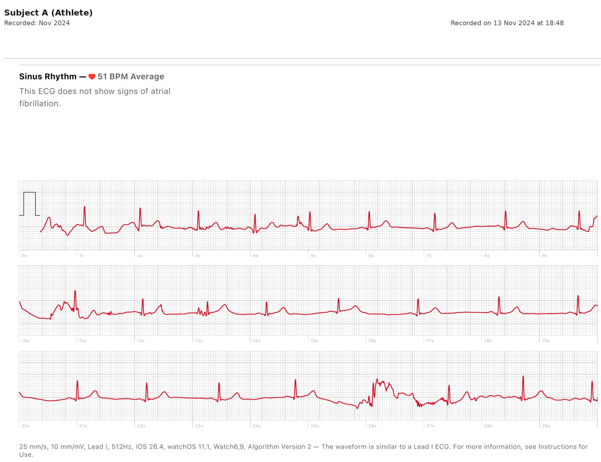

Fig. 7. Case 4a: Real Apple Watch ECG — Subject A (Athlete) (informed consent obtained). Sinus rhythm, 51 bpm, 512 Hz. Well-formed P-QRS-T complexes with consistent morphology across all three strips. Brief motion artefact at ~24 s (strip 3) does not affect classification. Apple Watch classification: "Sinus Rhythm — this ECG does not show signs of atrial fibrillation."

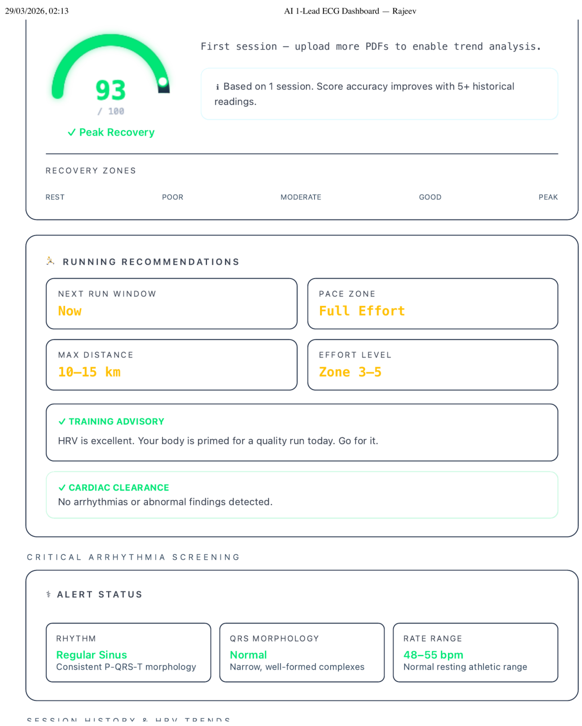

Fig. 8. Case 4b: Athlete dashboard output. Recovery score 93/100 (Peak). Training recommendation: Full Effort, Zone 3–5, 10–15 km. Cardiac Clearance: no arrhythmias detected. Rate range 48–55 bpm labelled "normal resting athletic range." At 51 bpm a general-population tool would generate a bradycardia false alarm — population-stratified thresholds prevent this. Real-world data; Apple Watch PDF input.

Summary: HRV Fingerprints Across Four Conditions

Case	HR	pNN50	Rate Range	Output
1 — VFib	199 bpm	90.9%	18–268 bpm	■ CRITICAL
2 — AFib	76 bpm	13%	73–80 bpm	■ DO NOT TRAIN
3 — SVT	171 bpm	2%	168–175 bpm	■ STOP
4 — Athlete NSR	51 bpm	24%	48–55 bpm	✓ PEAK RECOVERY

Three distinct tachyarrhythmias produce diagnostically separable HRV fingerprints. Paradoxically elevated RMSSD/pNN50 in VFib and depressed RMSSD/pNN50 in SVT both indicate pathology — confirming that HRV metrics must be interpreted in the context of rhythm validity, not in isolation.

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